MARCBEN JAMES SAMUEL S

Machine Learning & Data Science

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Summary

AI and Data Science student (CGPA 8.78) who takes models past the notebook: ML served behind APIs, LLM and speech models run offline, and models trained on heavily imbalanced data. First-author paper under review.

Education

St. Joseph's College of Engineering Chennai, India
B.Tech in Artificial Intelligence and Data Science; CGPA: 8.78/10 *Expected May 2028*
Coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, Probability and Statistics, DBMS

Technical Skills

ML and data: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy
LLMs, speech and vision: Hugging Face Transformers, RAG, LLM fine-tuning, Whisper, CLIP, OpenCV, ONNX
Engineering: FastAPI, Docker Compose, PostgreSQL, Redis, GitHub Actions CI, pytest, Git, Linux, Java, C

Experience

LGTG Solutions Pvt. Ltd. Remote
Backend Database / ML Engineering Intern *January 2026 – February 2026*
- Built an offline government voice assistant that answers citizen questions in Tamil, Hindi and English using a retrieval-augmented LLM pipeline, on a multilingual knowledge-base schema I designed and indexed.
- Profiled the serving path and cut average latency from 252 ms to 61 ms (76%) via retrieval and caching fixes.

Altascio Technologies On-site
Software / Data Engineering Intern *December 2025*
- Cleaned production datasets and engineered features for downstream machine-learning workflows.

IQmath Technologies LLP Remote
Data Analytics Intern *December 2025*
- Analysed multi-source business data with Python and SQL; built dashboards for non-technical stakeholders.

Selected Projects

SafeLobby — Fair Game-Chat Moderation *QLoRA, ONNX, FastAPI*
- Building a toxicity classifier for in-game chat by fine-tuning a small LLM with QLoRA on 45K Dota 2 chat lines (CONDA), with focal loss and validation-tuned thresholds for the rare toxic class.
- Auditing identity bias (subgroup, BPSN, BNSP AUC on Jigsaw) and serving an ONNX classifier that sends only borderline messages to the LLM, via FastAPI with CI/CD.

PhishGuard — Multi-Channel Phishing Detection *XGBoost, FastAPI, Redis* | github.com/DaganGera/PhishGuard
- Built an async FastAPI service that scores URLs, emails, QR codes and voice calls, deployed as a Docker Compose stack (PostgreSQL, Redis, MinIO) behind a WebSocket live threat dashboard and a Chrome extension.
- Fused an XGBoost model over 38 engineered URL features with VirusTotal reputation and WHOIS domain age, fetched concurrently, and added perceptual-hash screenshot matching to catch brand impersonation.
- Scored Whisper call transcripts for 7 social-engineering tactics; added Redis caching, rate limiting, API fallbacks and a retraining queue fed by user verdicts.

MediKiosk — Offline Multilingual AI Kiosk *LLM, Whisper, ONNX* | github.com/DaganGera/Patient-Case-Taking
- Built an on-premise clinical history-taking kiosk (SIH 2026) that runs Whisper ASR, a local Qwen2.5-1.5B LLM for schema-validated answer extraction and ONNX OCR for prescriptions, in 22 Indian languages, offline.
- GitHub Actions CI (lint, type checks, build) with 294 passing tests, including real-model tests.

SysHealth+ — Predictive Hardware Diagnostics *XGBoost, Ollama* | github.com/DaganGera/Sys-Health
- Labelled remaining useful life from per-drive SMART time series (about 10 GB of Backblaze logs), treating never-failed drives as censored; XGBoost failure model (under 1% positives) with class weights and PR-AUC.
- Shipped a live Windows monitor: background threads poll SMART via smartctl, flag deviations from each machine's own learned baseline, and an offline Mistral assistant explains faults from live telemetry.

Publications

Marcben James Samuel S, Raghul S, S. Sumathi, Sam V. George. *SPOT-TAP: Box-Supervised Few-Shot Defect-Type Classification with Frozen Foundation Models and a Support-Trained Patch Head.* Under review, 2026.

Achievements and Certifications

- **National Hackathon Winner**, Cyber Guardians '26 (Vel Tech); **Best Project Award**, New Prince College (2025); **President, Data Analytics Club**, organised Impact AI-Thon '26.
- **Microsoft Certified: Azure AI Fundamentals** (2026); Introduction to Machine Learning (NPTEL, Elite); Python for Data Science (NPTEL, Silver); Data Science Foundation (Coursera).